3D WEBSOFT INC. FOUNDATION

SCHOOL FUTURE TECH PROGRAM

Faculty Name :

Course Duration : 5 Months (150 Days)

Classes : Offline & Online

Daily Teaching : 3 Hours

Total Hours : 300 Hours

LEVEL 3 –Future Engineers(9th–10th Standard)

MODULE 1 – Python Foundations (Classes 1–10)

Class 7 – DataTypes

Complete Theory & Practical Notes

3DWebSoft Foundation

Learning Objectives

By the end of this session, the class aims to:

1. Introduce students to the concept of data in programming.
2. Explain why Python uses different data types.
3. Introduce the basic Python data types used by beginners.
4. Explain the difference between text, whole numbers, decimal numbers, and Boolean values.
5. Introduce str, int, float, and bool.
6. Teach students how to identify the data type of a value.
7. Introduce the type() function.
8. Demonstrate how data types can be stored in variables.
9. Help students select suitable data types for real-world information.
10. Provide hands-on practice creating and using different data types.
11. Develop students' ability to predict and verify the output of simple Python programs.
12. Build a foundation for future programming concepts such as input, operators, conditions, and calculations.

Session

Topic	Type
Warm-up: What is Data?	Discussion
Understanding Data Types	Theory
Strings – `str`	Theory + Demonstration
Integers – `int`	Theory + Demonstration
Floating-Point Numbers – `float`	Theory + Demonstration
Data Type Identification	Hands-on Practical 1
Break / Data Type Challenge	Activity
Boolean – `bool`	Theory + Demonstration
The `type()` Function	Theory + Practical
Student Profile Program	Hands-on Practical 2
Quiz + Exit Check	Assessment

3dWebSoft Foundation

1. Warm-up: What is Data?

Begin by asking students some simple questions:

- What is your name?
- How old are you?
- What is your height?
- What is your favorite subject?
- Are you a student?

Write example answers on the board:

```
Rahul
15
160.5
Science
True
```

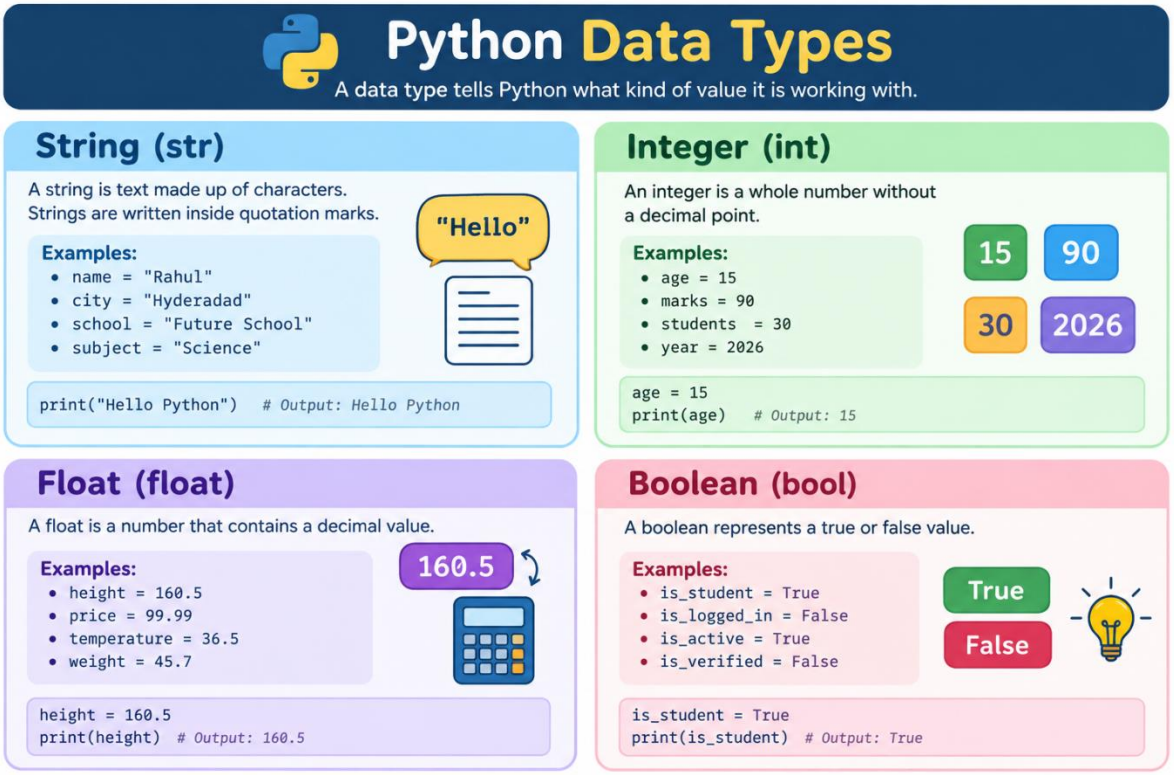
Data is information that a computer program can store, process, and use.

Examples:

```
Name        → "Rahul"
Age         → 15
Height      → 160.5
Student     → True
```

Introduce the main idea:

A data type tells Python what kind of value it is working with.



2. Understanding Data Types

THEORY

What is a Data Type?

A **data type** identifies the kind of value stored or used by a Python program.

For beginners, focus on four important Python data types:

```
str    → String
```

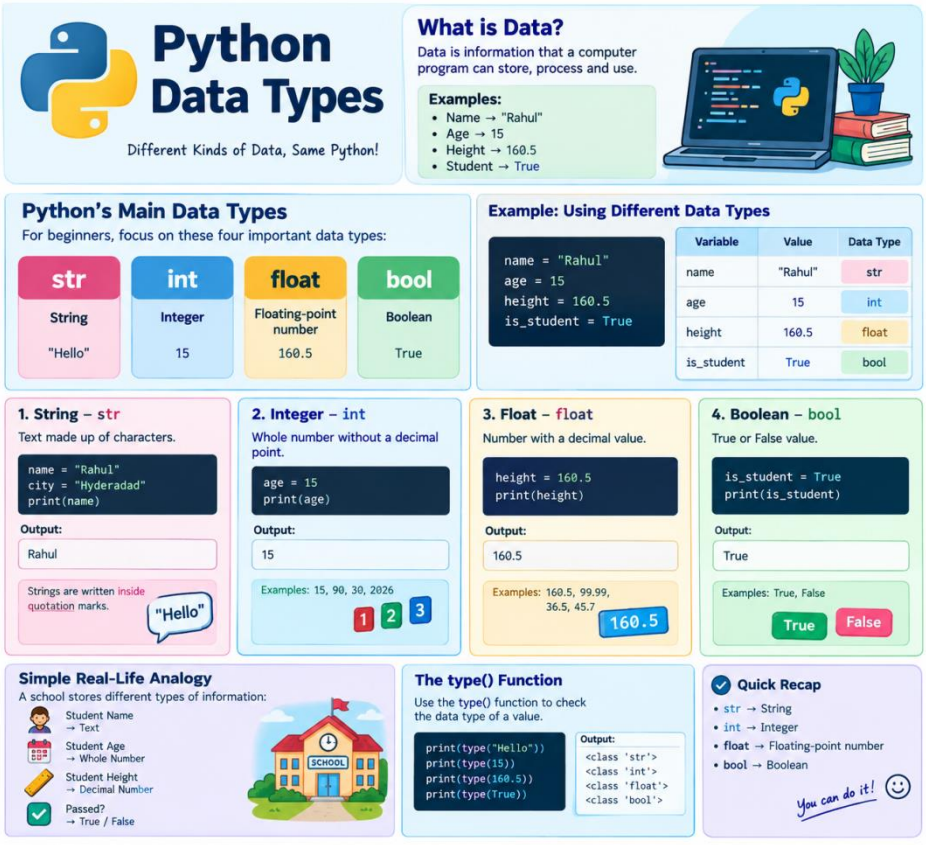
```
int   → Integer
float → Floating-point number
bool  → Boolean
```

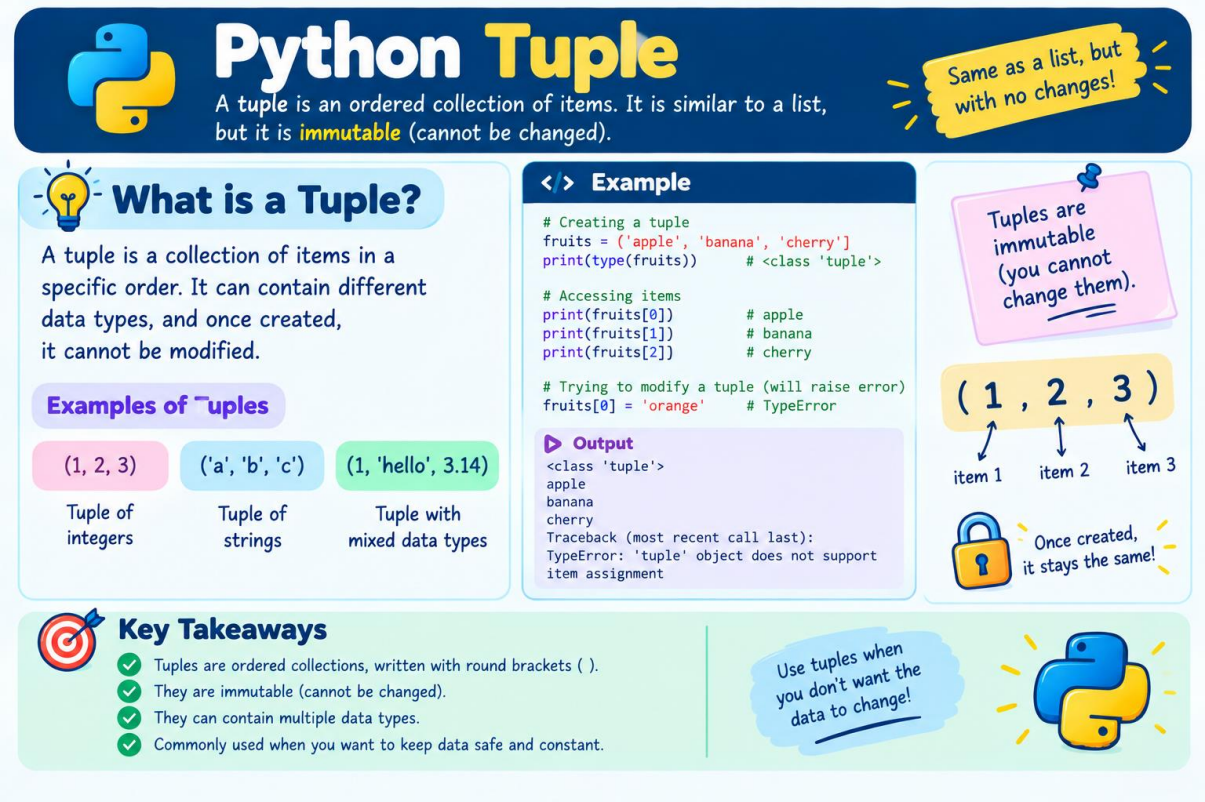
Example

```
name = "Rahul"
age = 15
height = 160.5
is_student = True
```

Variable	Value	Data Type
name	"Rahul"	str
age	15	int
height	160.5	float
is_student	True	bool

3d ft Foundation





Simple Real-Life Analogy

Explain that a school stores different types of information:

```
Student Name    → Text
Student Age     → Whole Number
Student Height  → Decimal Number
Passed?         → True / False
```

Python also needs to distinguish between different kinds of information.

3. String – str

THEORY + DEMONSTRATION

What is a String?

A **string** is text made up of characters.

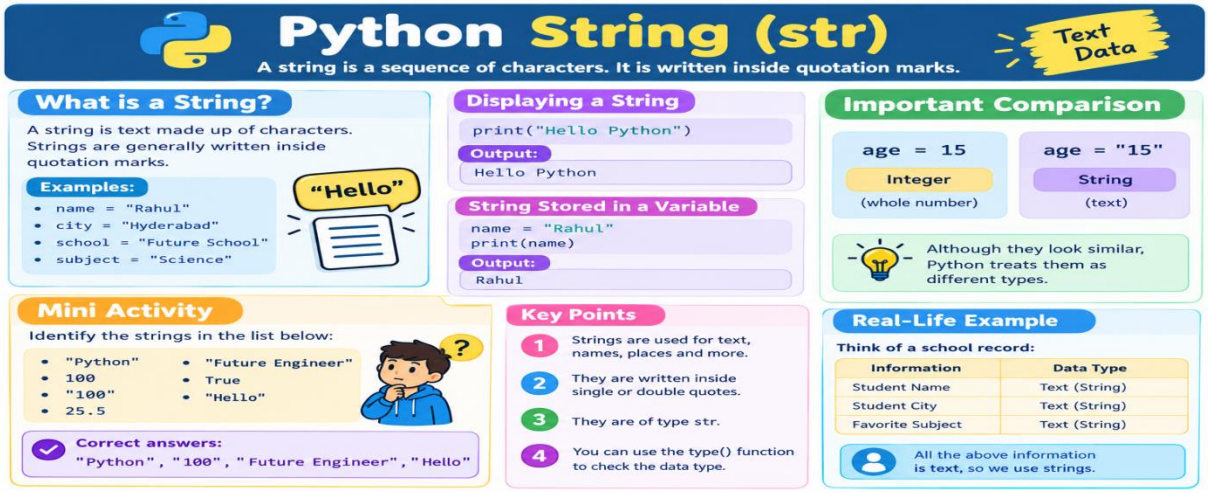
Strings are generally written inside quotation marks.

Examples:

3dWebSoft Fo

```
name = "Rahul"
city = "Hyderabad"
school = "Future School"
subject = "Science"
```

All of these are strings.



Displaying a String

```
print("Hello Python")
```

Output:

```
Hello Python
```

String Stored in a Variable

```
name = "Rahul"
print(name)
```

Output:

```
Rahul
```

Important Comparison

Show students:

```
age = 15
```

versus:

3dWebSoft Fo

```
age = "15"
```

Explain:

```
15   → integer
"15" → string
```

Although they look similar, Python treats them as different types.

Mini Activity

Ask students to identify the strings:

```
"Python"
100
"100"
25.5
"Future Engineer"
True
```

Correct answers:

```
"Python"
"100"
"Future Engineer"
```

4. Integer – int

THEORY + DEMONSTRATION

What is an Integer?

An **integer** is a whole number without a decimal point.

Examples:

```
age = 15
marks = 90
students = 30
year = 2026
```

Example

```
age = 15
print(age)
```

Output:

```
15
```

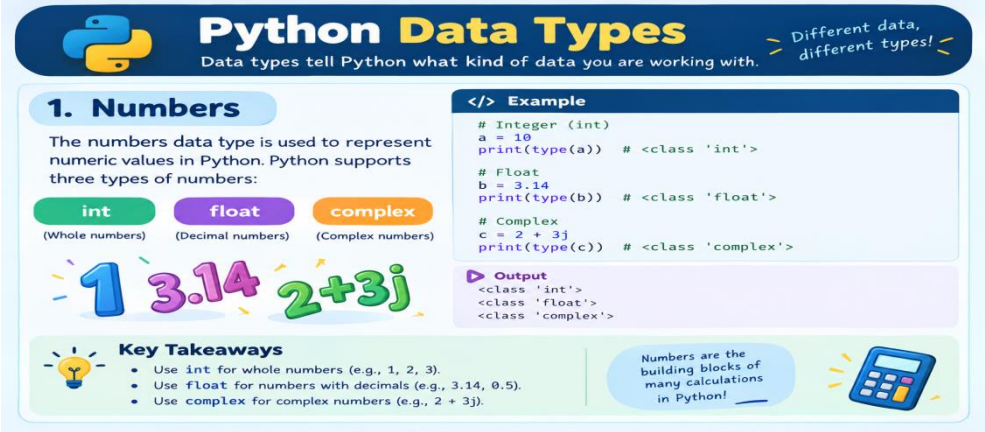
3dWebSoft Foundation

Integer Examples

```
score = 95
number_of_students = 35
books = 10
```

Explain:

When we need a whole number, we can use an integer.



5. Float – float

THEORY + DEMONSTRATION

What is a Float?

A **float** is a number that contains a decimal value.

Examples:

```
height = 160.5
price = 99.99
temperature = 36.5
weight = 45.7
```

Example

```
height = 160.5

print(height)
```

Output:

```
160.5
```

3WebSoft

Integer vs Float

```
age = 15
height = 160.5
```

Value	Type
15	int
160.5	float

Explain:

Integer = whole number
Float = decimal number

6. Data Type Identification

HANDS-ON PRACTICAL 1

Students create a file:

```
datatypes_practice.py
```

Write:

```
name = "Anita"
age = 14
height = 155.5
school = "Future School"
marks = 85
```

Ask students to complete:

Variable	Value	Data Type
name	"Anita"	?
age	14	?
height	155.5	?
school	"Future School"	?
marks	85	?

Expected Answers

3dWebSoft Foundation

```
name   → str
age    → int
height → float
school → str
marks  → int
```

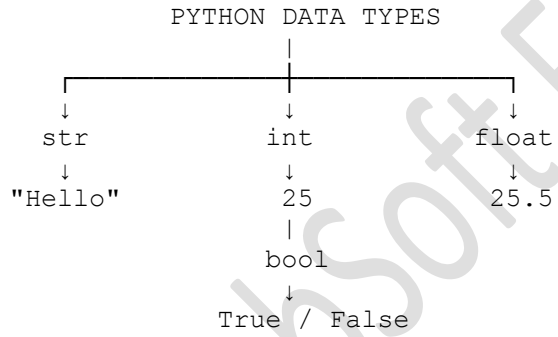
Students then run:

```
print(name)
print(age)
print(height)
print(school)
print(marks)
```

Suitable Images – Data Types

Recommended classroom visual

Use a four-section visual:



7. Break / Data Type Challenge

Give students these values:

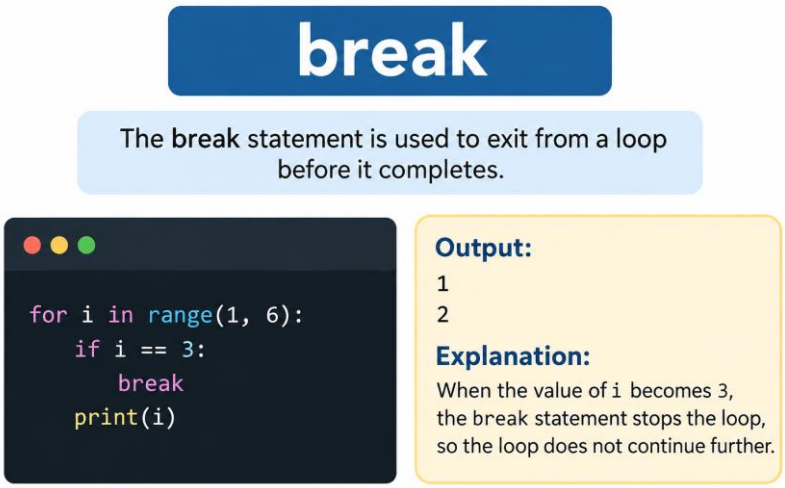
```
"Python"
50
75.5
True
"100"
False
25
"Engineer"
```

Create four categories:

```
STRING | INTEGER | FLOAT | BOOLEAN
```

3dWebSoft Foundation

Students classify each value.



Pair Activity

Student A reads a value.

Student B identifies the data type.

Example:

Student A: 75.5
Student B: float

Then students swap roles.

8. Boolean – bool

THEORY + DEMONSTRATION

What is Boolean Data?

A Boolean represents one of two possible values:

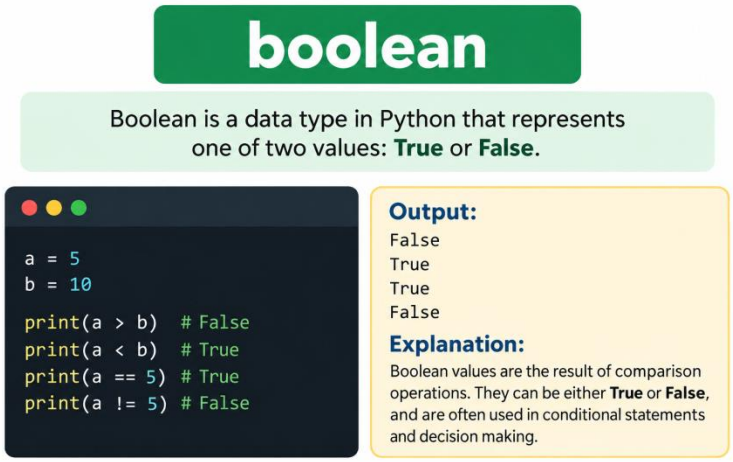
```
True
False
```

Boolean values are useful when something has two possible states.

Examples

3dWebSoft Foundation

```
is_student = True
is_raining = False
has_ticket = True
```



Real-Life Examples

Question	Possible Boolean
Are you a student?	`True` / `False`
Is the light on?	`True` / `False`
Did the student pass?	`True` / `False`

Example Program

```
is_student = True

print(is_student)
```

Output:

```
True
```

Important

Python uses:

```
True
False
```

The first letters are capitalized.

3dWebSoft Foundation

9. The type() Function

THEORY + PRACTICAL

Introduce the built-in type() function.

Explain:

type() tells us the data type of a value or variable.

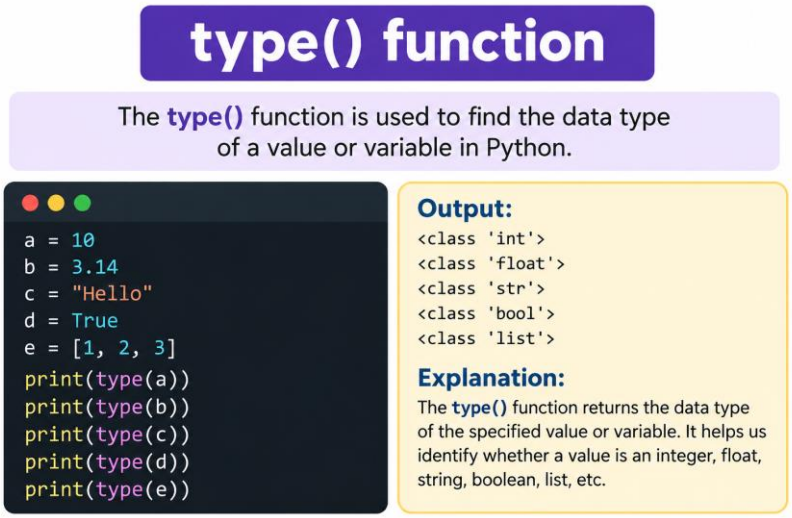
Example – String

```
name = "Rahul"

print(type(name))
```

Output:

```
<class 'str'>
```



Example – Integer

```
age = 15

print(type(age))
```

Output:

```
<class 'int'>
```

Example – Float

```
height = 160.5

print(type(height))
```

Output:

```
<class 'float'>
```

Example – Boolean

```
is_student = True

print(type(is_student))
```

Output:

```
<class 'bool'>
```

Practical Activity – "Type Detective"

Students write:

```
name = "Anita"
age = 14
height = 155.5
is_student = True

print(type(name))
print(type(age))
print(type(height))
print(type(is_student))
```

Prediction Activity

Before running the program, students write down what they think the output will be.

Then they run the program and compare their prediction with the actual result.

This helps develop **logical thinking and prediction skills**.

10. Student Profile Program

HANDS-ON PRACTICAL 2

3dWebSoft Foundation

Students create:

`student_profile.Py`

Starter Code

```
name = "Rahul"
age = 15
height = 160.5
is_student = True
favorite_subject = "Science"

print("===== STUDENT PROFILE =====")
print("Name:", name)
print("Age:", age)
print("Height:", height)
print("Student:", is_student)
print("Favorite Subject:", favorite_subject)
```

Expected Output

```
===== STUDENT PROFILE =====
Name: Rahul
Age: 15
Height: 160.5
Student: True
Favorite Subject: Science
```

Personalization Task

Students replace the values with their own information.

For example:

```
name = "________"
age = 14
height = 150.5
is_student = True
favorite_subject = "________"
```

3dWebSoft Foundation

.



11. Final Data Type Challenge

Program containing **all four basic data types**.

Example:

```
name = "Anita"
age = 14
height = 155.5
is_student = True

print(name)
print(age)
print(height)
print(is_student)
```

Then ask students to add:

```
print(type(name))
print(type(age))
print(type(height))
print(type(is_student))
```

Challenge Questions

1. Which variable contains a string?
2. Which variable contains an integer?
3. Which variable contains a float?
4. Which variable contains a Boolean?
5. What will type(age) return?

12. Quick Assessment

Question 1

What is a data type?

Answer: A data type identifies the kind of value used in a program.

Question 2

What is the data type of:

```
"Hello"
```

Answer: str

Question 3

What is the data type of:

```
25
```

Answer: int

Question 4

What is the data type of:

```
25.5
```

Answer: float

Question 5

What is the data type of:

```
True
```

Answer: `bool`

Question 6

3dWebSoft Foundation

Which function is used to find the data type?

Answer:

```
type()
```

13. Classroom Game – "Data Type Sorting"

Prepare cards containing:

```
"Python"
100
25.5
True
"Engineer"
50
99.9
False
"100"
```

Create four groups:

```
STRING
INTEGER
FLOAT
BOOLEAN
```

Extension Question

Why is "100" a string while 100 is an integer?

Expected response:

"100" is written as text inside quotation marks, while 100 is a number.

Learning Outcomes

By the end of this class, students will be able to:

1. **Explain** what data and data types mean in programming.
2. **Identify** the basic Python data types str, int, float, and bool.
3. **Recognize** and use strings for storing text.
4. **Identify and use** integers for whole-number values.
5. **Identify and use** floats for decimal-number values.
6. **Identify and use** Boolean values such as True and False.
7. **Create** variables containing different Python data types.

3dWebSoft Foundation

8. **Use** print() to display values stored in variables.
9. **Use** the type() function to determine the type of a value or variable.
10. **Differentiate** between values such as 100 and "100".
11. **Select** an appropriate data type for simple real-world information.
12. **Predict and verify** the data type and output of simple Python statements.
13. **Create and run** a Python program containing multiple data types.
14. **Interpret** the results produced by the type() function.
15. **Apply** data type concepts to create a simple Student Profile program.

3dWebSoft Foundation